Diffusion-Augmented Contrastive Learning in Low-Data Histopathology Regimes

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Abstract

Deep learning models in computational pathology encounter severe performance deterioration when target biopsy classes have limited annotated samples. We introduce Diff-AugCLR, a framework combining conditional latent diffusion models with self-supervised contrastive learning to synthesize diverse cellular morphology and reinforce representation learning. Evaluated across two diverse histopathology benchmarks (Camelyon16 and TCGA-COAD), our framework demonstrates substantial robustness improvements in low-label regimes.

1. Introduction

Histopathologic classification of whole-slide images (WSIs) forms the gold standard for oncologic staging. However, annotating gigapixel slides requires substantial pathologist time, leading to severe label scarcity for uncommon neoplastic subtypes. While standard geometric transformations provide basic data augmentation, they cannot synthesize diverse nuclear pleomorphism, chromatin distribution, or glandular architectural variation.

2. Methods and Multi-Dataset Benchmarks

We implemented a two-stage pipeline: a class-conditional latent diffusion model trained on tissue patches, followed by MoCo-v3 contrastive pre-training using both real and synthesized histological patches.

Datasets:

1. Camelyon16: 400 whole-slide images for breast cancer sentinel lymph node metastasis detection. Tissue patches extracted at 20x magnification (0.5 um/pixel).
2. TCGA-COAD: The Cancer Genome Atlas colorectal adenocarcinoma cohort containing 460 patients. Patches extracted across 9 morphological tissue categories including tumor, stroma, and normal mucosa.

Experimental Evaluation Regimes:

To test label efficiency, models were fine-tuned using stratified subsets of 5%, 10%, and 25% labeled training slides, evaluated against fully supervised baselines.

3. Comparative Findings Across Datasets
Across both Camelyon16 and TCGA datasets, the technique consistently improved classification accuracy by an average margin of 5.8% in low-data regimes.

Specific Metric Gains:
On Camelyon16 with only 10% labeled data, the baseline ResNet-50 attained an AUC of 0.812, whereas Diff-AugCLR achieved an AUC of 0.887 (+7.5% absolute gain). On the multi-class TCGA-COAD benchmark with 10% labels, balanced accuracy increased from 74.3% to 80.9%.

Plateau in High-Data Regimes:
Importantly, when labeled training data exceeded 50% of available slides, diffusion augmentation yielded no statistically significant improvement over standard affine transformations (AUC 0.941 vs 0.939, $p = 0.42$). In high-data regimes, real sample diversity already saturates model capacity.

4. Limitations and Architectural Boundaries
Our experimental findings reveal several key limitations:

1. Hallucination Risk: In approximately 2.3% of generated high-grade tumor tiles, the diffusion model synthesized non-biological nuclear configurations that do not correspond to valid histological criteria.

2. Hardware and Training Cost: Training the high-resolution conditional diffusion model required 4x NVIDIA A100 GPUs for 72 hours, consuming significant electrical power.

3. Patch-Level Confinement: The current framework operates on localized 512x512 patches and does not model organ-level macroscopic tissue architecture or vascular invasion patterns spanning multiple millimeters.